Ex 4: OFDMA Resource Block Allocation

AIM: To simulate and analyze OFDMA (Orthogonal Frequency Division Multiple Access)resource block allocation and evaluate system performance in terms of resource utilization and user throughput using MATLAB.

```
clc; clear; close all;

% System parameters

bandwidth = 10e6; % 10 MHz

rb_size = 180e3; % 180 kHz per RB

% Total Resource Blocks

total_RB = floor(bandwidth / rb_size);

% Users

users = 1:6;

% RB allocation (simple equal + remainder distribution)

base = floor(total_RB / length(users));


alloc = base * ones(1,length(users));

alloc(1:mod(total_RB,length(users))) = alloc(1:mod(total_RB,length(users))) +
1;

utilization = (alloc / total_RB) * 100;

disp('Total Resource Blocks:')

disp(total_RB)

disp('RB Allocation per User:')

disp(alloc)

figure

% -------- Graph 1: RB Allocation per User --------

subplot(2,1,1)

stem(users, alloc, 'filled', 'LineWidth', 2)
```

title('OFDMA Resource Block Allocation per User')

xlabel('User Index')

ylabel('Number of Resource Blocks')

grid on

% -------- Graph 2: RB Utilization --------

subplot(2,1,2)

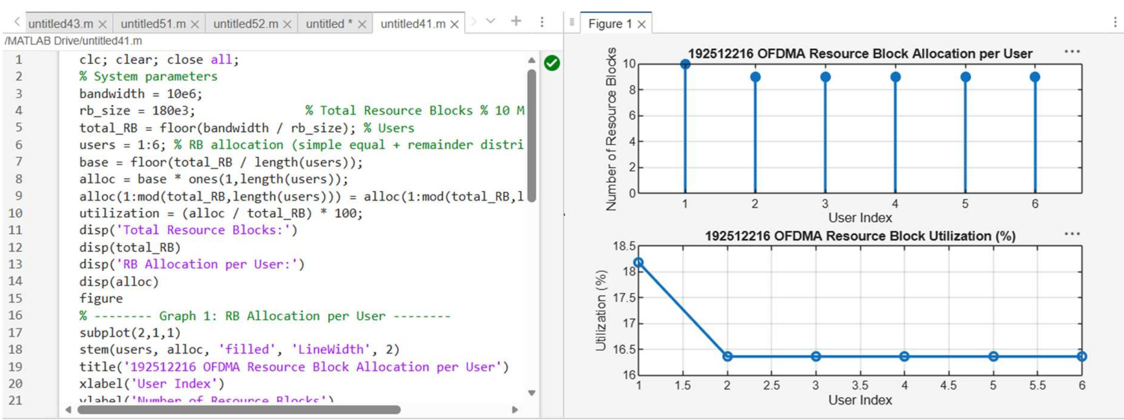
plot(users, utilization, '-o', 'LineWidth', 2)

title('OFDMA Resource Block Utilization (%)')

xlabel('User Index')

ylabel('Utilization (%)')

grid on



Objective 2 Matlab Code and Output:

clc; clear; close all;

% Total system resource blocks

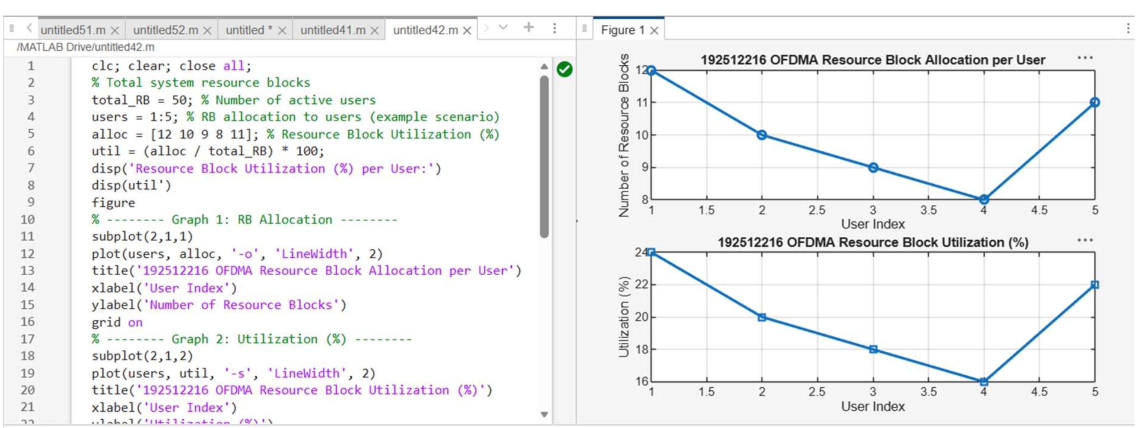
total_RB = 50;

% Number of active users

users = 1:5;

% RB allocation to users (example scenario)

```
alloc = [12 10 9 8 11];
% Resource Block Utilization (%)
util = (alloc / total_RB) * 100;
disp('Resource Block Utilization (%) per User:')
disp(util')
figure
% -------- Graph 1: RB Allocation --------
subplot(2,1,1)
plot(users, alloc, '-o', 'LineWidth', 2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on
% -------- Graph 2: Utilization (%) --------
subplot(2,1,2)
plot(users, util, '-s', 'LineWidth', 2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on
```

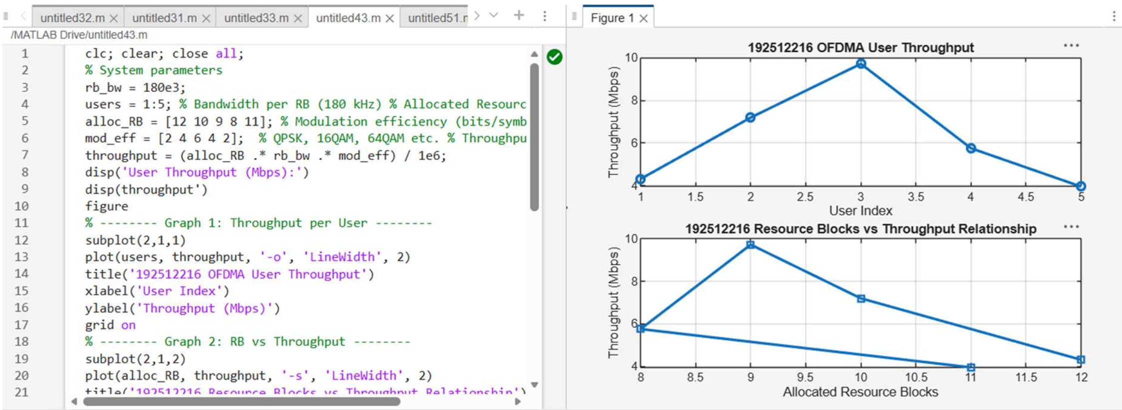


Objective 3 Matlab Code and Output:

```
clc; clear; close all;

% System parameters

rb_bw = 180e3; % Bandwidth per RB (180 kHz)

users = 1:5;

% Allocated Resource Blocks per user

alloc_RB = [12 10 9 8 11];

% Modulation efficiency (bits/symbol)

mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.

% Throughput calculation (Mbps)

throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;

disp('User Throughput (Mbps):')

disp(throughput')

figure

% -------- Graph 1: Throughput per User --------

subplot(2,1,1)

plot(users, throughput, '-o', 'LineWidth', 2)

title('OFDMA User Throughput')

xlabel('User Index')

ylabel('Throughput (Mbps)')

grid on

% -------- Graph 2: RB vs Throughput --------

subplot(2,1,2)

plot(alloc_RB, throughput, '-s', 'LineWidth', 2)

title('Resource Blocks vs Throughput Relationship')

xlabel('Allocated Resource Blocks')
```

ylabel('Throughput (Mbps)')

grid on



Result:

1. The Resource Block (RB) allocation was successfully simulated for multiple users in the OFDMA system, demonstrating how available spectrum is distributed among active users.

2. The Resource Block Utilization (%) analysis showed efficient usage of system resources, indicating how effectively the available RBs are assigned to users.

3. The User Throughput (Mbps) was successfully calculated based on RB allocation and modulation efficiency, showing that higher allocated RBs and higher modulation schemes result in increased throughput.